

The empirical recurrence for OEIS A236742: recovering a conjecture that is not written down

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Abstract

OEIS A236742 records “Empirical recurrence of order 86”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 657$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 86, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 86 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A236742 is “Number of $(n+1) \times (4+1)$ 0..2 arrays with the upper median plus the lower median of every 2×2 subblock equal.”. It has offset 1 and begins

2907, 46377, 810039, 17740369, 385709603, 8588436417, 191076235235, 4264088666817, ...

The entry states:

Empirical recurrence of order 86 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 09:31:16, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 657$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 657$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 1314$ exact terms of the model returns order 86 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{86} c_i a(n-i),$$

c_1	56	c_{23}	−88446472645102617151	c_{45}	−11078842104635443948380076365
c_2	−971	c_{24}	189703042852733857249	c_{46}	31468475919075686426485822419
c_3	−152	c_{25}	832480918929966534951	c_{47}	12354722441315253154852427645
c_4	190873	c_{26}	−3147931526385388195938	c_{48}	−76519317526070181580388021284
c_5	−1661887	c_{27}	−4982867947477027772269	c_{49}	6133192123440808923868375827
c_6	−10009206	c_{28}	35744713444977948535612	c_{50}	147531434786839346425260796567
c_7	213828146	c_{29}	6602349093447677536009	c_{51}	−67607820835288691508451652323
c_8	−312175856	c_{30}	−301209810911293867384168	c_{52}	−225963172752204105161432620551
c_9	−12610205483	c_{31}	238699555746690517592636	c_{53}	185450411753089956307334138380
c_{10}	66713398446	c_{32}	1917867056603366392212474	c_{54}	270558574911593990653527754082
c_{11}	384188036780	c_{33}	−3250879857566599449169865	c_{55}	−338725158984611952959386745108
c_{12}	−4069754544077	c_{34}	−9021390986329552113938280	c_{56}	−241102032962494027581075625768
c_{13}	−2970246585544	c_{35}	25702420173974928592296374	c_{57}	467511584304256743355102024856
c_{14}	145450214494923	c_{36}	28130870271634328062490137	c_{58}	135477959681789037882763204724
c_{15}	−260085066899947	c_{37}	−147062988088131091098368746	c_{59}	−507016704763296591995520520240
c_{16}	−3389529900369997	c_{38}	−26482957345743932638765509	c_{60}	−1807913563854425524833517816
c_{17}	13437100535257914	c_{39}	645125392261628761703284995	c_{61}	437526343153511336076569994464
c_{18}	49901573197783878	c_{40}	−304952979906921662062790849	c_{62}	−95174312359171190384206308032
c_{19}	−364155800303484757	c_{41}	−2202780261306128925726760480	c_{63}	−300348193561033103745352744768
c_{20}	−309048914527576159	c_{42}	2329867427067214688914531910	c_{64}	123182847888821401799137972864
c_{21}	6693706416724637849	c_{43}	5787794445026478112504831387	c_{65}	162396678032868024066180336384
c_{22}	−5293405206431529293	c_{44}	−10041438664471031213750311635	c_{66}	−97810180468021909967610445824

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 86, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $86 + 4$ terms removed returns order 86 as well, so $\deg q_{\min} = 86 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once

more on the sequence with its first $86 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 86 that this sequence satisfies — and that family turns out to have exactly one member.

References

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